

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: HOPE | Context: US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is single-label classification per utterance [Q14, Q30] with macro-F1, weighted-F1, and accuracy metrics [Q63] plus label-wise F1 reporting [Q76]. Statistical significance via T-test at >95% confidence is reported [Q88], and 3-fold cross-validation results are consistent with split-based results [Q87]. The output modality (categorical labels over text utterances) directly matches the deployment's expected output for session quality assessment and counselor feedback. Macro-F1 weights all classes equally regardless of frequency, appropriate for the imbalanced label distribution. The user marks OF as LOWER priority (form is well matched). Minor concerns: macro-F1 does not surface per-label safety-critical performance directly (e.g., crisis-relevant IRQ/YNQ behavior is averaged in); error analysis [Q89, Q90] is on in-distribution data only and does not assess behavior on out-of-scope (peer/crisis) utterances. Score of 4 reflects strong form alignment with minor diagnostic limitations.

ID	Evidence
Q14	"Each utterance in the dataset is tagged with one of the 12 counseling-aligned dialogue-act labels (c.f. Section 3)." (p.2)
Q30	"Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance." (p.3)
Q63	"To measure the performances of SPARTA and other baseline systems, we compute macro-F1, weighted-F1, and accuracy scores." (p.6)
Q76	"We also present the label-wise performance of SPARTA in Table 4." (p.7)
Q87	"Moreover, we also report the mean of the 3-fold cross-validation results for both SPARTA-MHA and SPARTA-TAA, and the results are consistent with the train-val-test split case." (p.7)
Q88	"We also perform statistical significance T-test comparing SPARTA-TAA and the best performing baseline (CASA). We observe that our results are significant with > 95% confidence across macro-F1 (p-value= 0.009), weighted-F1 (p-value= 0.014), and accuracy (p-value= 0.048) values." (p.7)
Q89	"Error Analysis: In this section, we present two-way error analyses of SPARTA in terms of quantitative and qualitative evaluations." (p.7)
Q90	"Quantitative analysis: We report the confusion matrix for SPARTA-TAA in Figure 4." (p.7)

Strengths

- Output modality (categorical labels) directly matches deployment use case for flow classification [Q14, Q63]
- Multiple complementary metrics reported (macro-F1, weighted-F1, accuracy) [Q63] including per-label F1 [Q76]
- Statistical significance testing [Q88] and 3-fold cross-validation [Q87] support reliability claims
- Quantitative and qualitative error analysis [Q89, Q90, Q91] provides interpretable failure-mode information

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Q90	"Quantitative analysis: We report the confusion matrix for SPARTA-TAA in Figure 4." (p.7)
Q91	"We observe three pairs with significant error-rates ($\geq 25\%$) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%)." (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — categorical-label output over text utterances matches deployment requirements (session flow classification for feedback/supervision).

OF-2: Not applicable — deployment is text-only; no TTS requirement indicated in deployment context.

OF-3: Not applicable as a primary concern — practitioners (not therapy clients) are the system's users; literacy is high in the practitioner population.

OF-4: Minor gaps: no separate evaluation of safety-critical label categories; no out-of-distribution evaluation framework for peer/crisis utterances; macro-F1 averaging may mask label-specific failure modes that matter for clinical review.

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Information Gaps

- No reported evaluation framework for safety-critical label subsets
- No reported out-of-distribution behavior on peer/crisis-register utterances

Requires Expert Verification

- Whether macro-F1 at the reported 60.29 threshold is sufficient for supervisory feedback in clinical training contexts

Remediation

Gap 1: Macro-F1 averaging masks safety-critical label performance; no out-of-distribution evaluation for peer/crisis utterances [Q63, Q76].

Recommendation: Report and monitor per-label F1 (especially IRQ, YNQ, ACK, GC) separately in deployment dashboards. Add an out-of-distribution detection layer that flags utterances likely outside the trained label distribution rather than forcing a label assignment, particularly for crisis-context deployment.

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